Ainekko MXM Board

Modified 03/09/2026

aifoundry@nekko.ai

AiNEKKO.

Doc Title: MXM			Doc Num: 400-00050		
Page Title: 01 - Cover Sheet			Drawn By: Bruce Klein		
Date: Monday, March 09, 2026	Size: C		Sheet: 1 of 32		Rev: 1

Voltage Rail Definitions and Ranges

Voltage Rail	Boot Voltage	PMIC Supported Range	Where Used
VDD_MNN	500mV	305mV - 750mV	Minions and some Neighborhood
VDD_NOC	430mV	305mV - 750mV	NOCs
VDD_SCW	750mV	550mV - 900mV	SCW logic including L2/L3 SRAM, 4x4 crossbar, Neighborhood 32KB shared L1 caches/logic, level shifters, LVPLLs, Sprinkler
VDD_MXN	850mV	650mV - 885mV	4 Maxions and their caches
VDD_DDR	800mV	700mV - 950mV	DRAM controllers (synopsys IP for 1066MHz) in MemShire
VDD_LOGIC	775mV	700mV - 900mV	Logic in IO shires, GPIO, PShire, fuse blocks, DFT
VDD_Q	1.1V	1.06V - 1.17V	DRAM internals, SoC/DRAM PHYs
VDD_QLP	640mV	530mV - 630mV	IO voltage to DRAM and SoC
VDD_PCIE_IO	1.5V	1.4V - 1.8V	PCIe IO
P1_8VSTDBY	1.8V	1.8V	PMIC Controller
VDDA_1P8	1.8V	1.8V	Analog 1.8V
VDD_1P8	1.8V	1.8V	SoC GPIO voltages, random board logic
P3_3V	3.3V	3.3V	USB and random board components

Note: Boot Voltages based upon SoC FW 0.8.0, PMIC FW 0.7.0
Values may change. Consult the firmware release documents.

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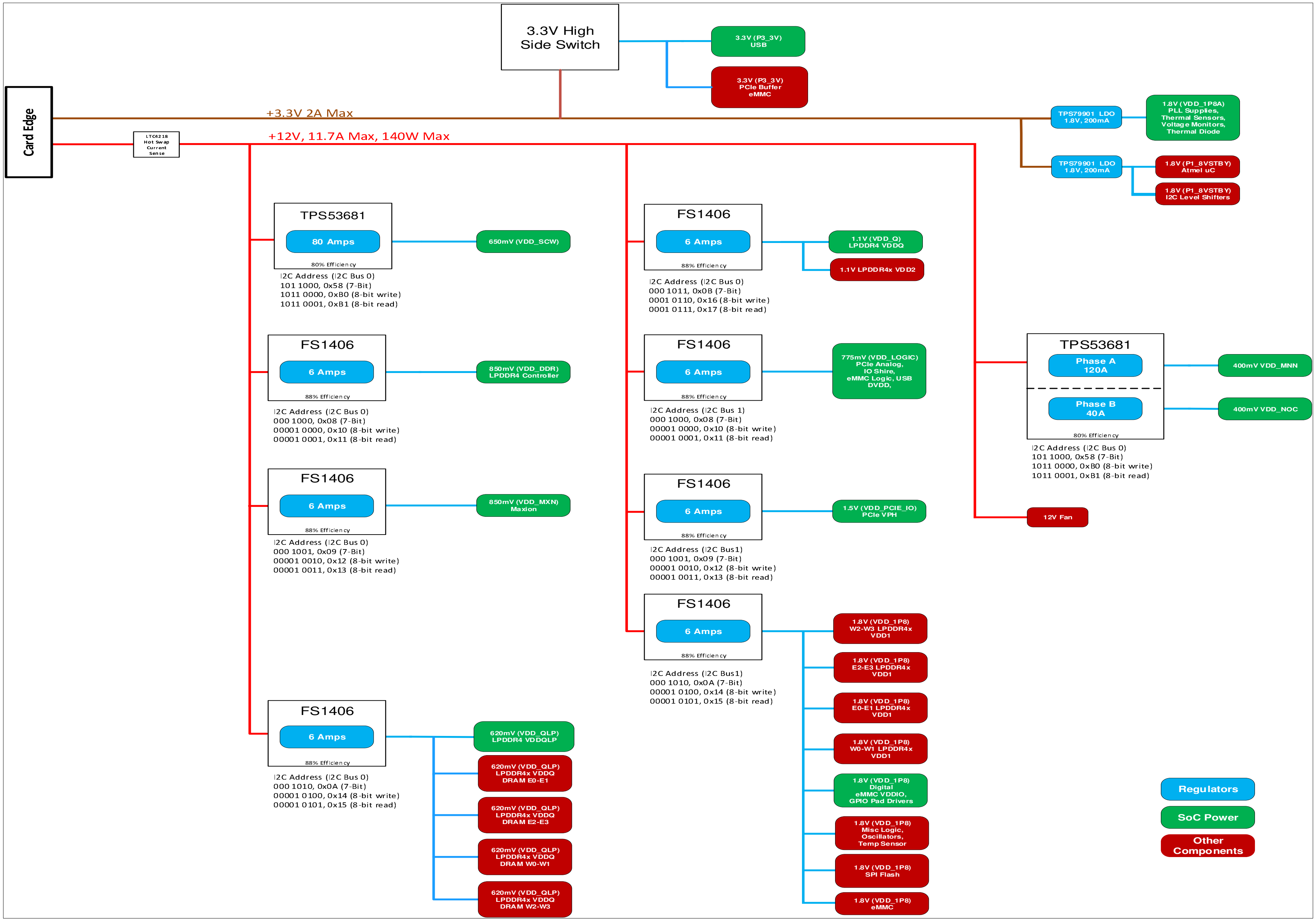

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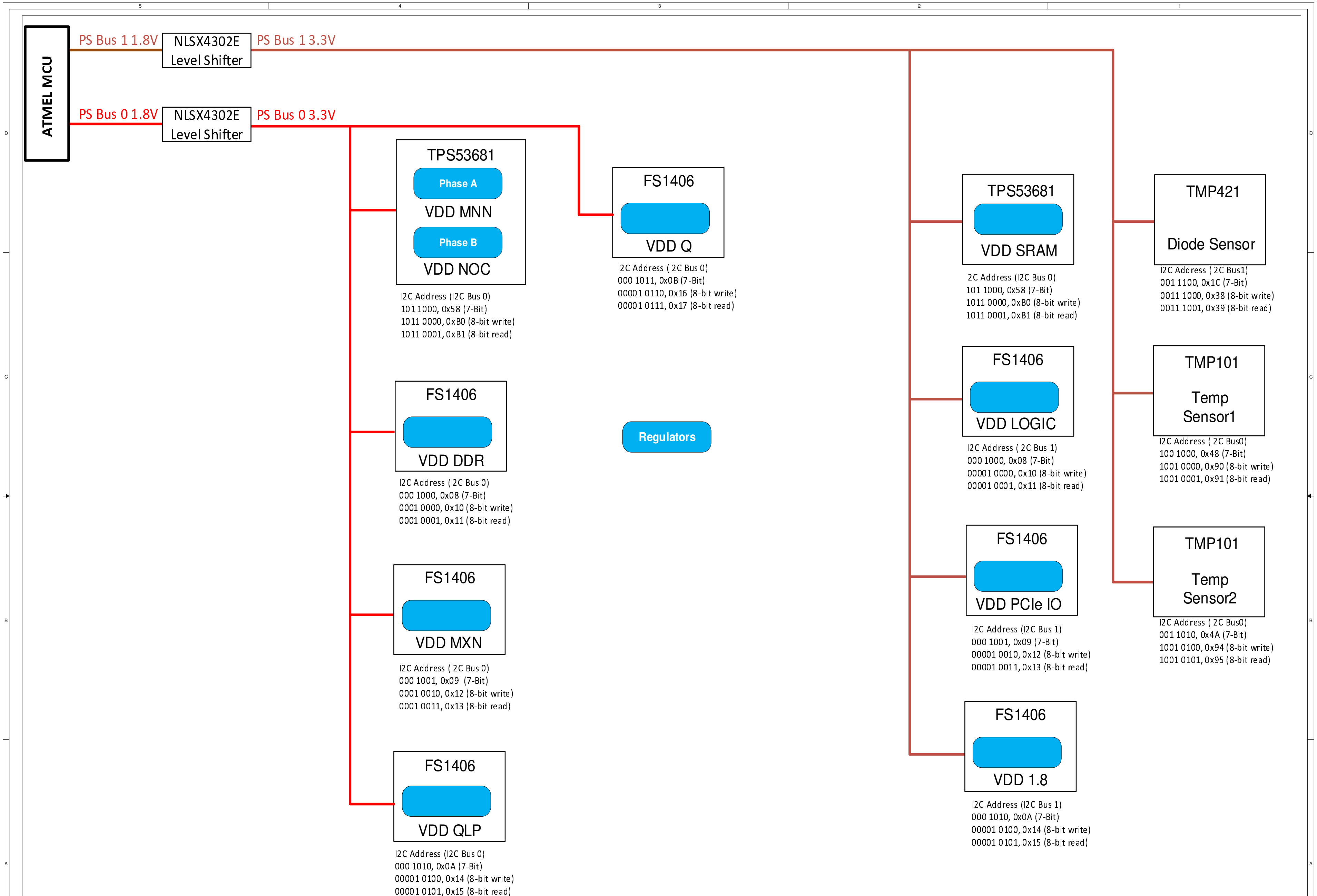
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 03 - Power Tree		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 3 of 32	Rev:



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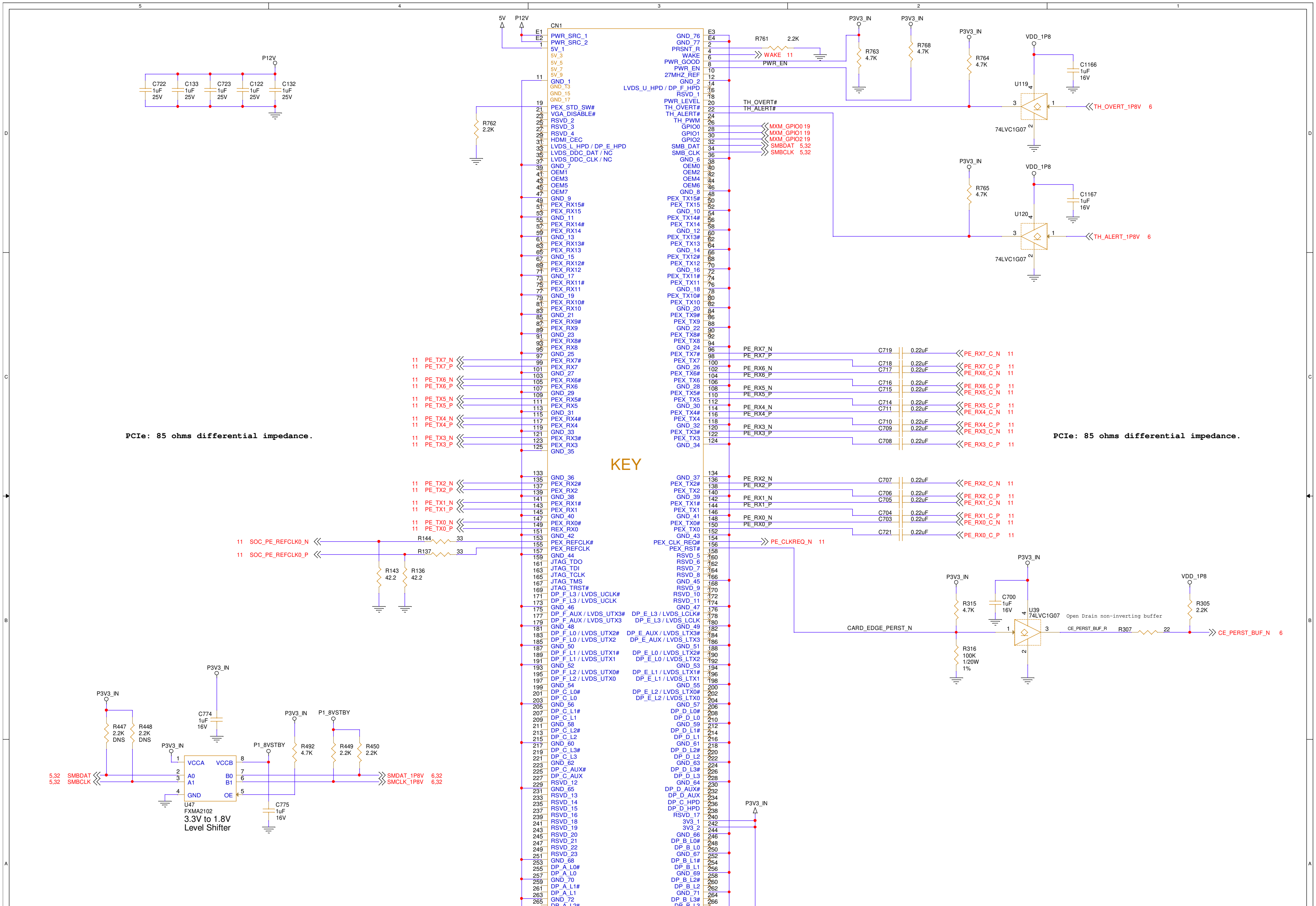
Doc Title: MXM			Doc Num: 400-00050		
Page Title: 04 - I2C Buses			Drawn By: Bruce Klein		
Date: Monday, March 09, 2026	Size: C		Sheet: 4 of 32		Rev:



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Doc Title: MXM

Doc Num: 400-00050

Page Title: 05 - MXM Card Edge Connector

Drawn By: Bruce Klein

Date: Monday, March 09, 2026

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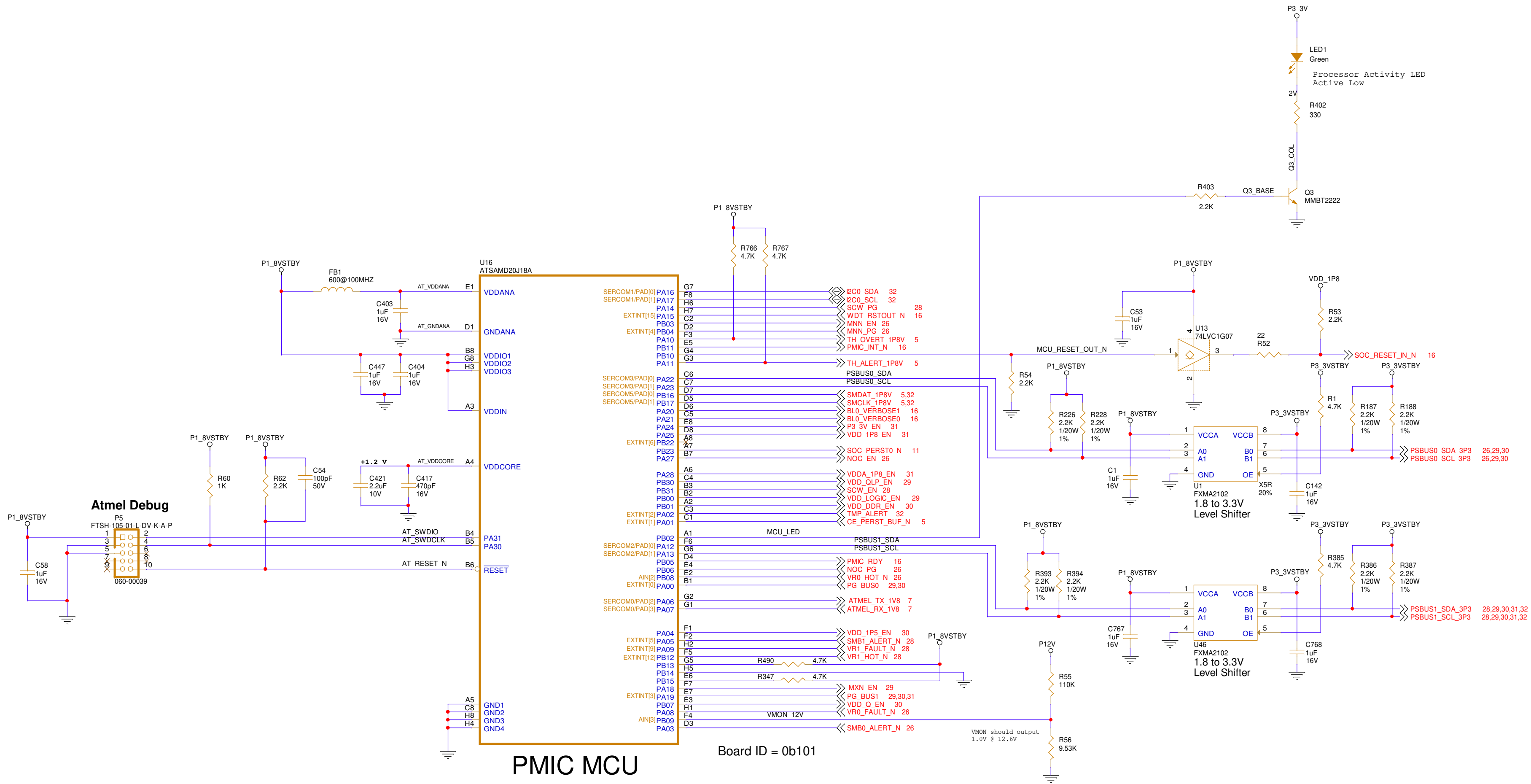

Signal Name	Type	Description
PAD[0]	Digital I/O	SDA
PAD[1]	Digital I/O	SCL
PAD[2]	Digital I/O	SDA_OUT (4-wire operation)
PAD[3]	Digital I/O	SCL_OUT (4-wire operation)

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Doc Title: MXM

Doc Num: 400-00050

Page Title: 06 - Atmel MCU

Drawn By: Bruce Klein

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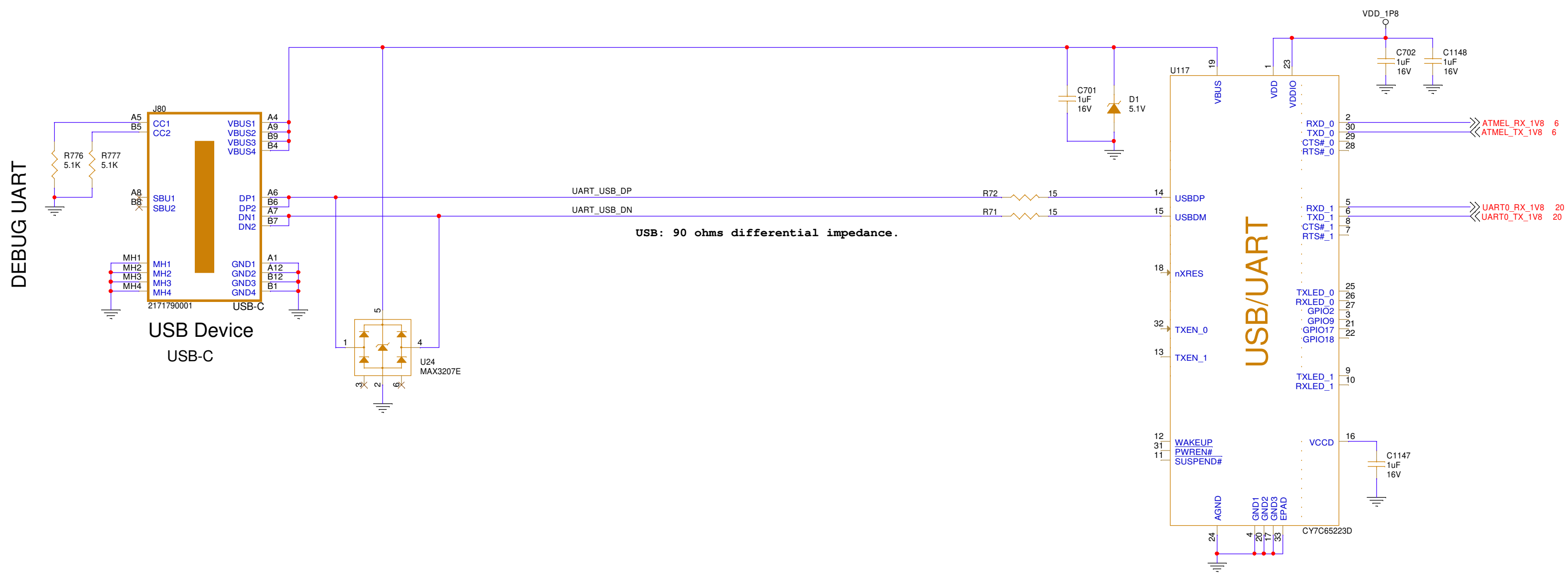

UART is not active until VDD_1P8
is enabled to prevent leakage
current to SoC

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 07 - USB-to-Serial Converter		Drawn By: Bruce Klein	
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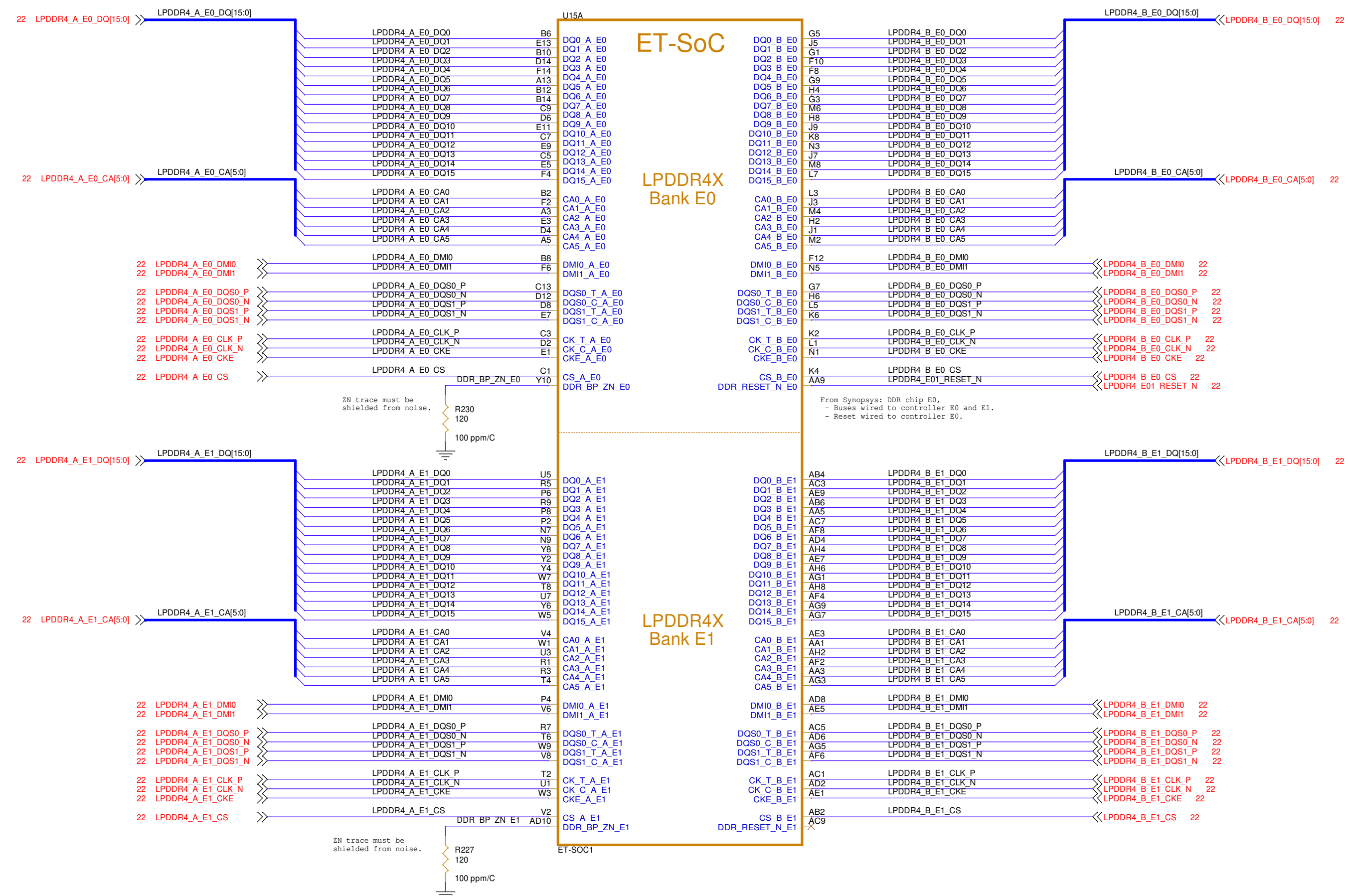

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Doc Title: MXM
Doc Num: 400-00050
Page Title: 08 - SOC - LPDDR4x - E01
Drawn By: Bruce Klein
Date: Monday, March 09, 2026
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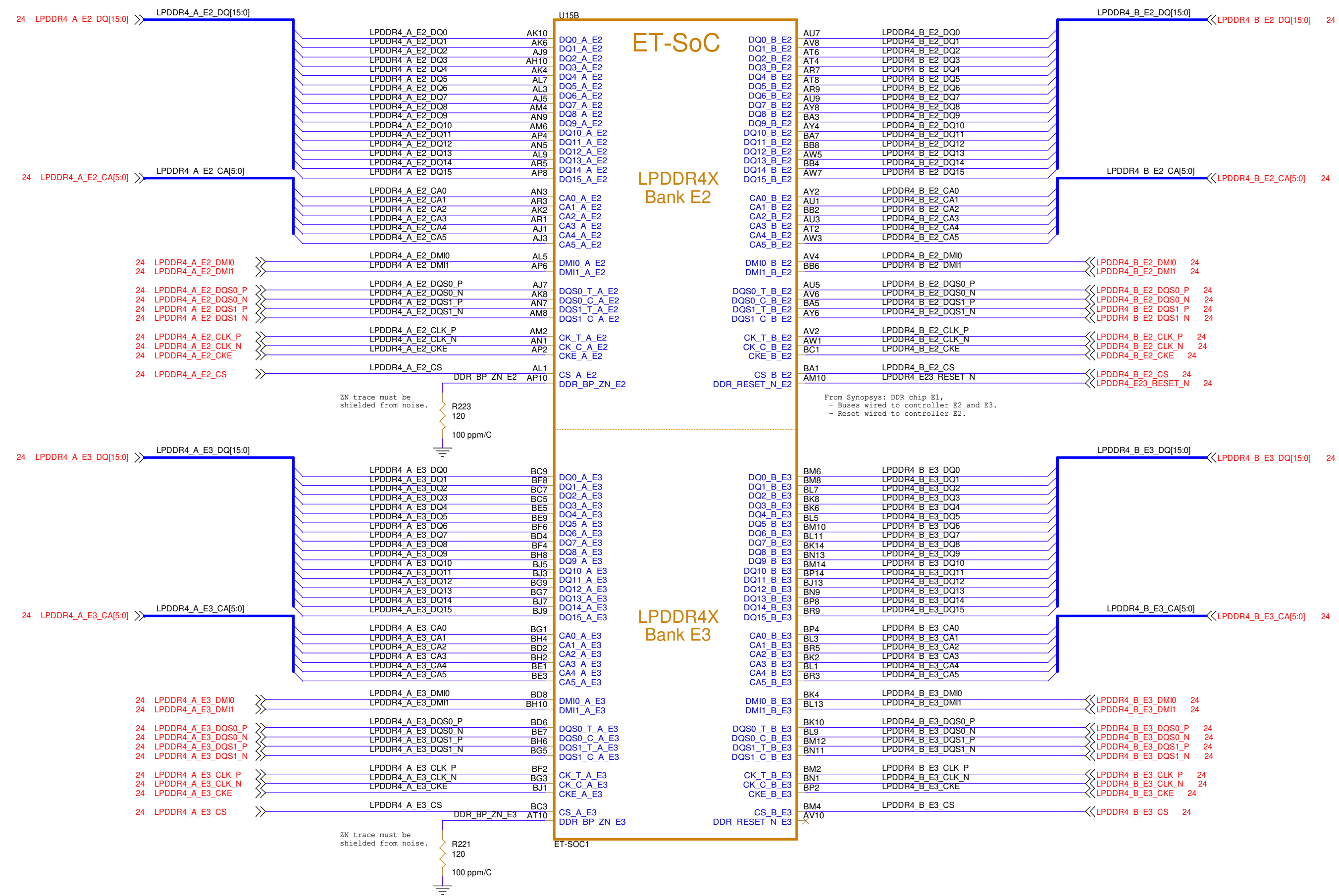

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 09 - SOC - LPDDR4x - E23		Drawn By: Bruce Klein	
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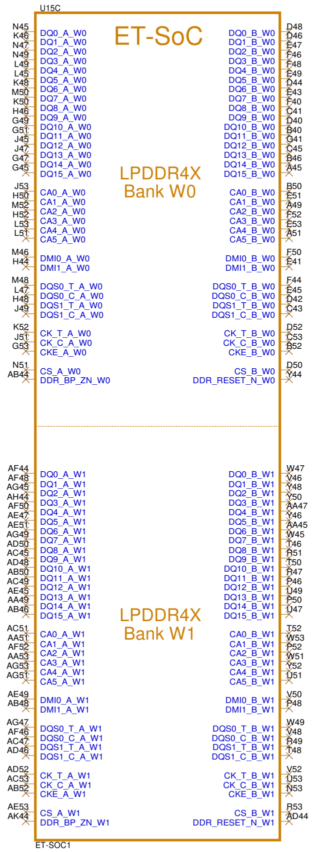

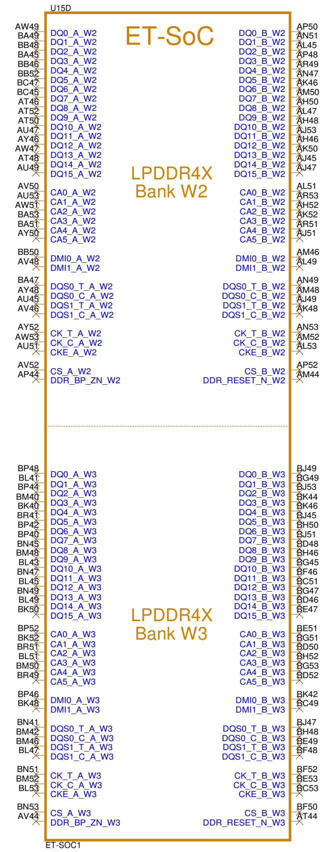

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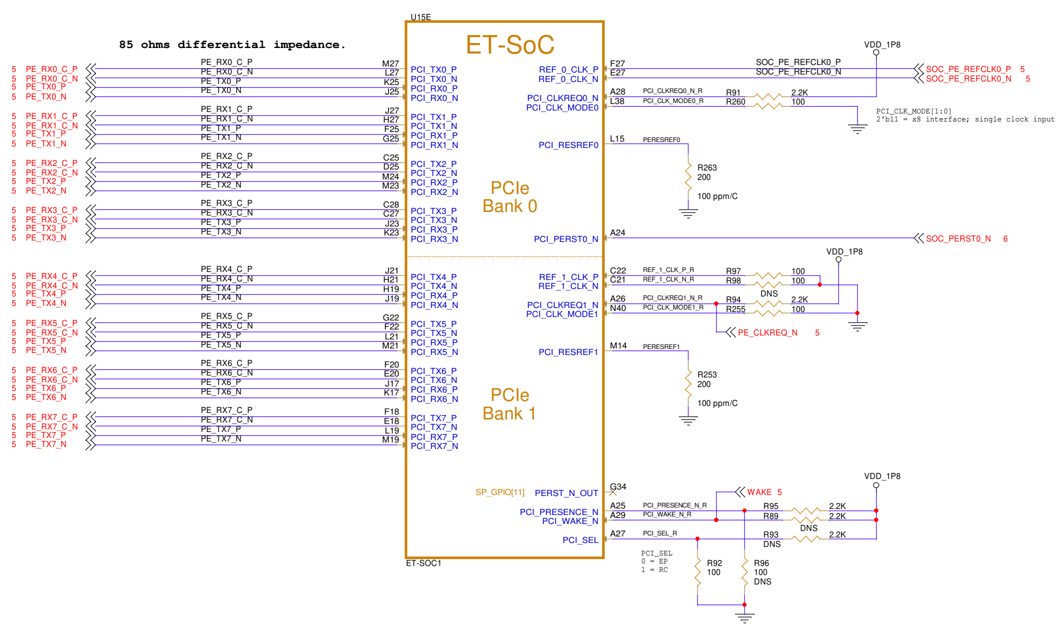
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 10 - SOC - LPDDR4x - WEST		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 10 of 32	Rev: 1

Esperanto PCISIG ID is x1E0A.



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Doc Title: MXM

Doc Num: 400-00050

Page Title: 11 - SOC - PCIe

Drawn By: Bruce Klein

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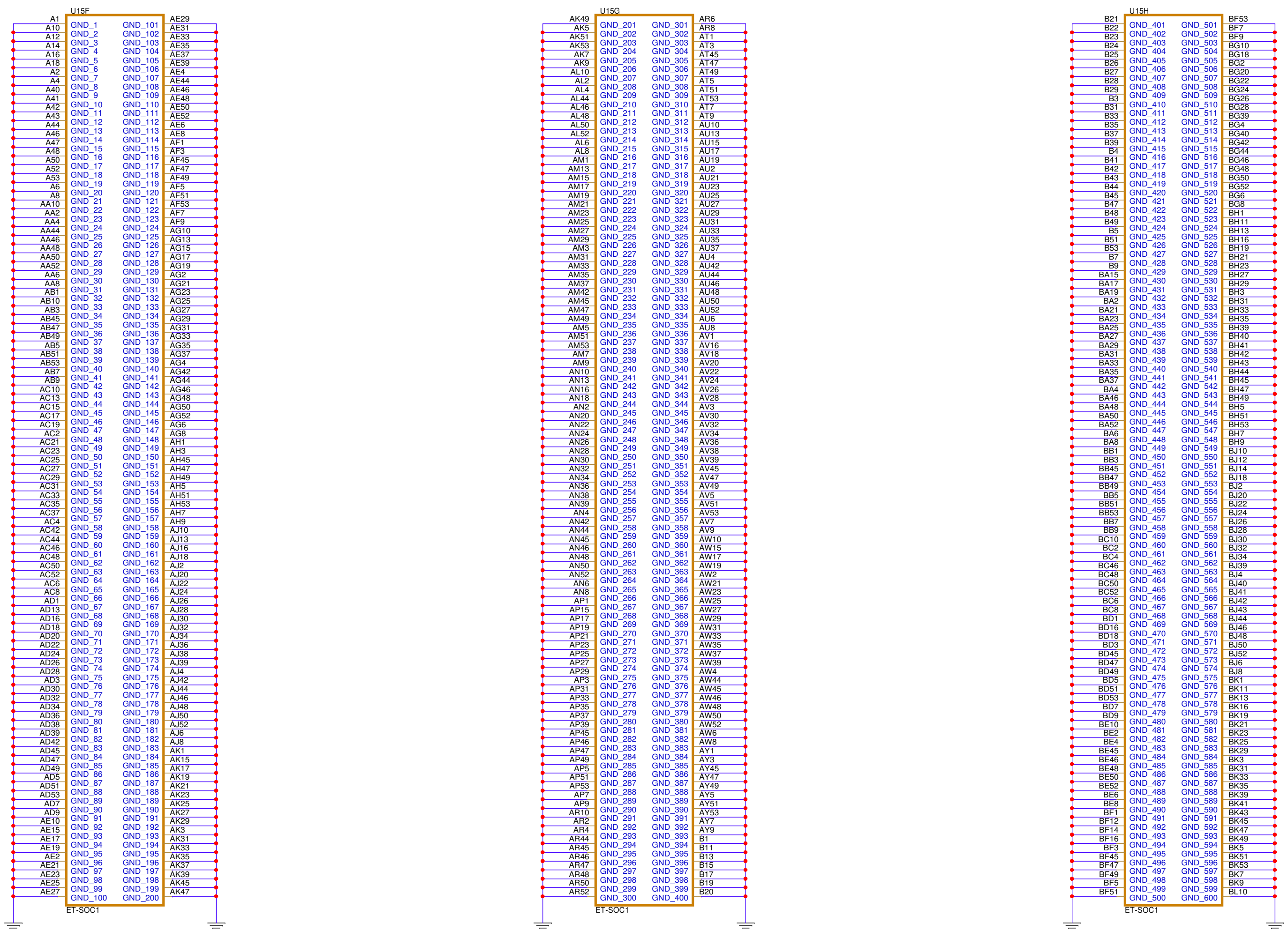

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Doc Title: MXM

Doc Num: 400-00050

Page Title: 12 - SOC - GND 1

Drawn By: Bruce Klein

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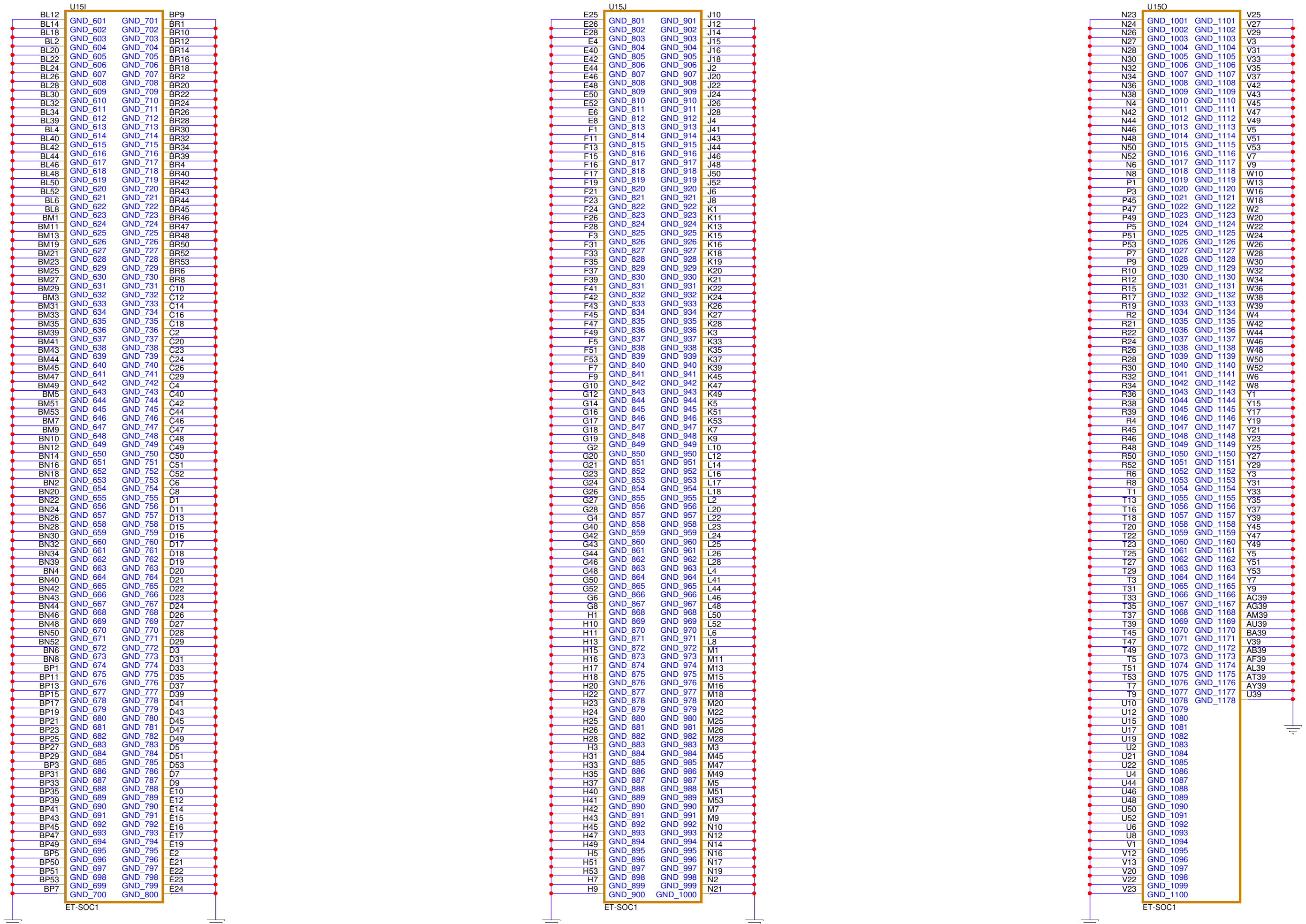

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Doc Title: MXM

Doc Num: 400-00050

Page Title: 13 - SOC - GND 2

Drawn By: Bruce Klein

Date: Monday, March 09, 2026

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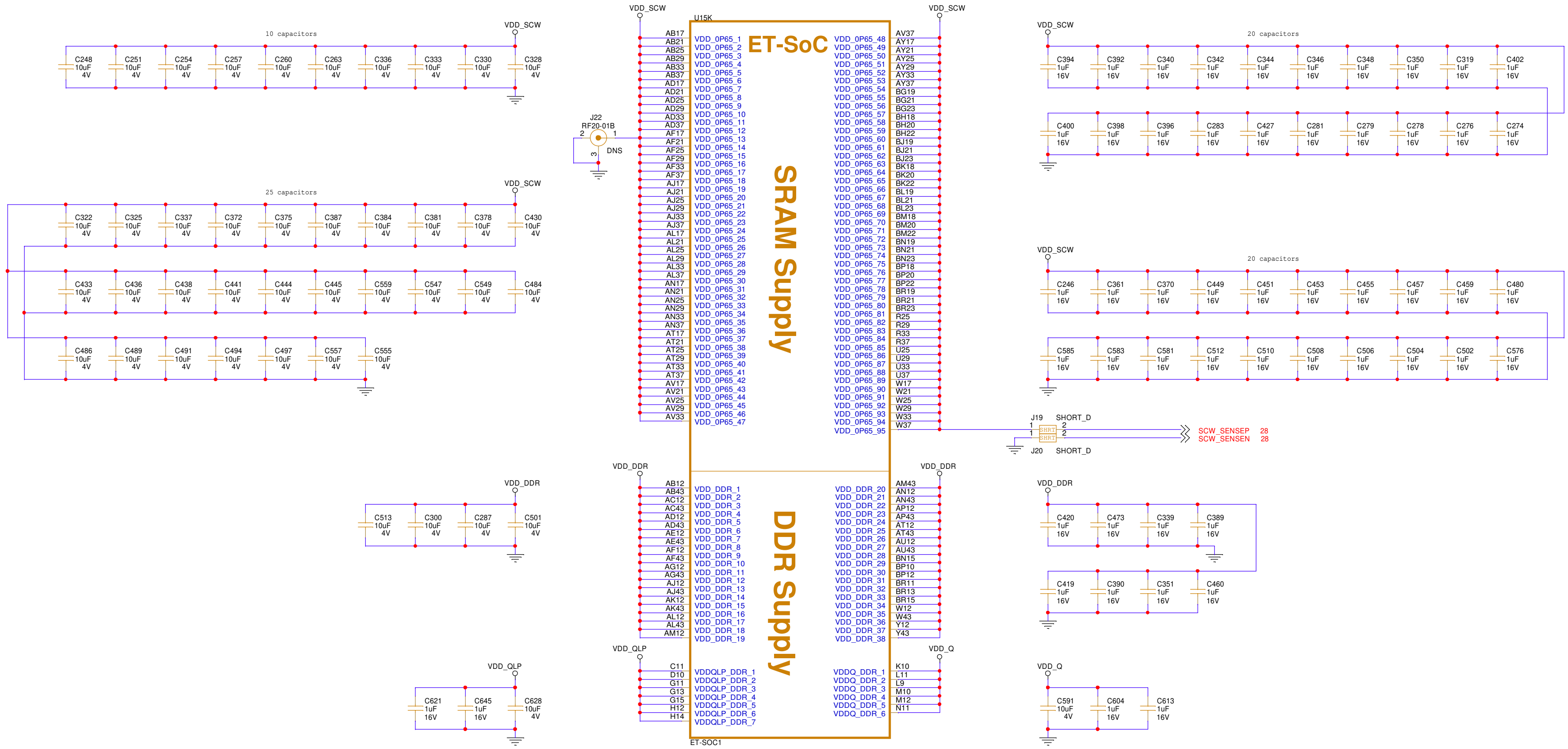
Rev: 1



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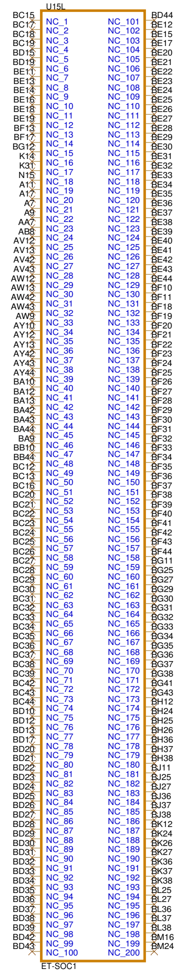
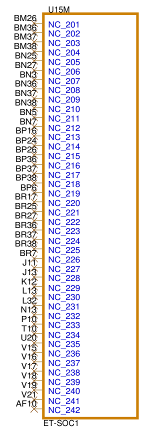
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 14 - SOC - DRAM and SRAM Power		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 14 of 32	Rev: 1





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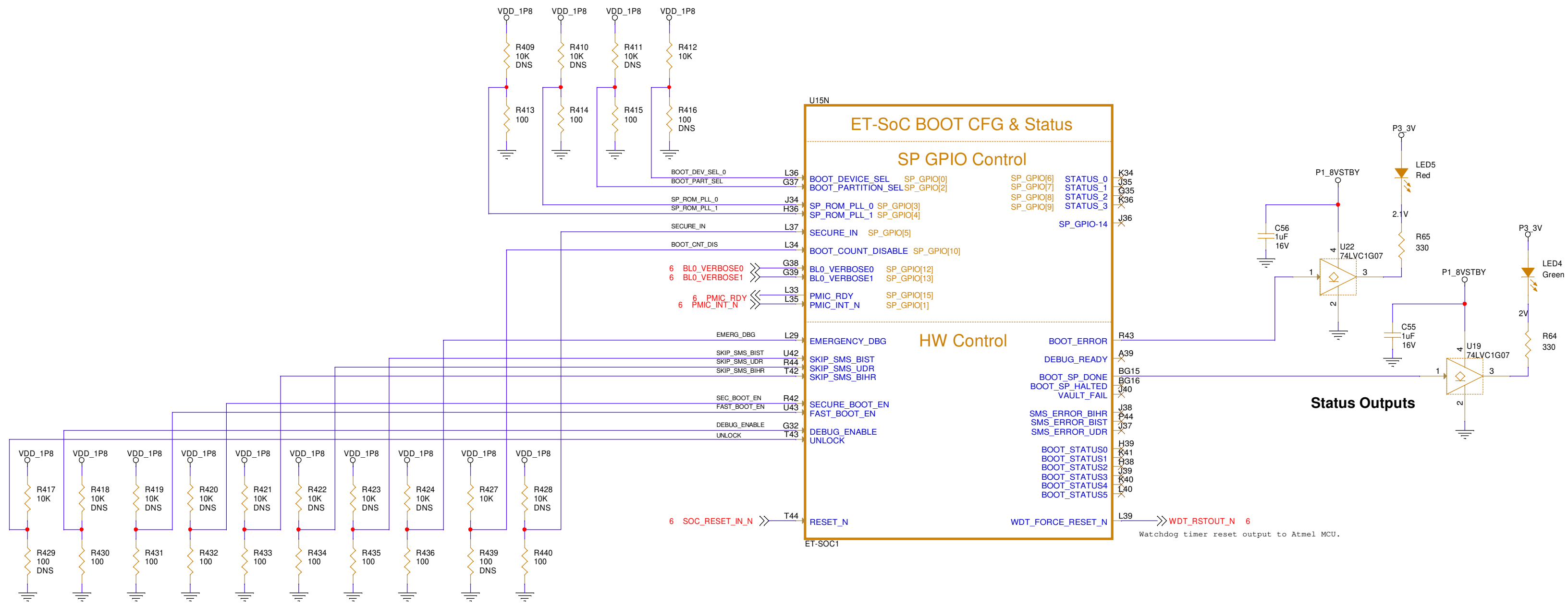
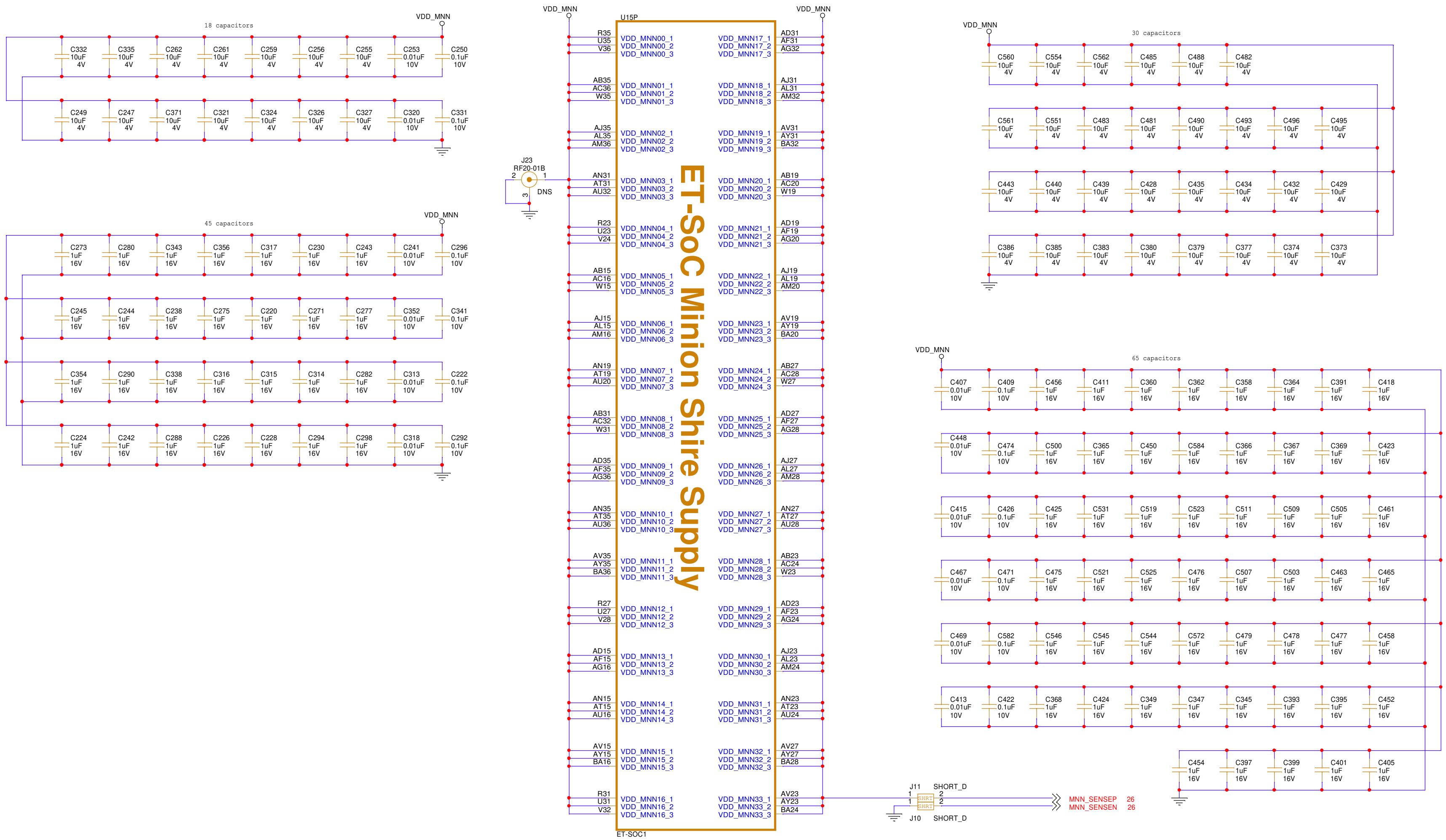
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 15 - SOC - NC Pins		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 15 of 32	Rev: 1



SOC - Pin Strap Inputs

Pin Strap	Default Value
BOOT_DEVICE_SEL	High
BOOT_PART_SEL	Low
SP_ROM_PLL_0	Low
SP_ROM_PLL_1	Low
SECURE_IN	Low
BL0_VERBOSE0	PMIC
BL0_VERBOSE1	PMIC
EMERGENCY_DBG	Low
SKIP_SMS_BIST	Low
SKIP_SMS_UDR	Low
SKIP_SMS_BIHR	Low
SECURE_BOOT_EN	Low
FAST_BOOT_EN	Low
DEBUG_ENABLE	Low
UNLOCK	High



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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 17 - SOC - Minion Power		Drawn By: Bruce Klein	
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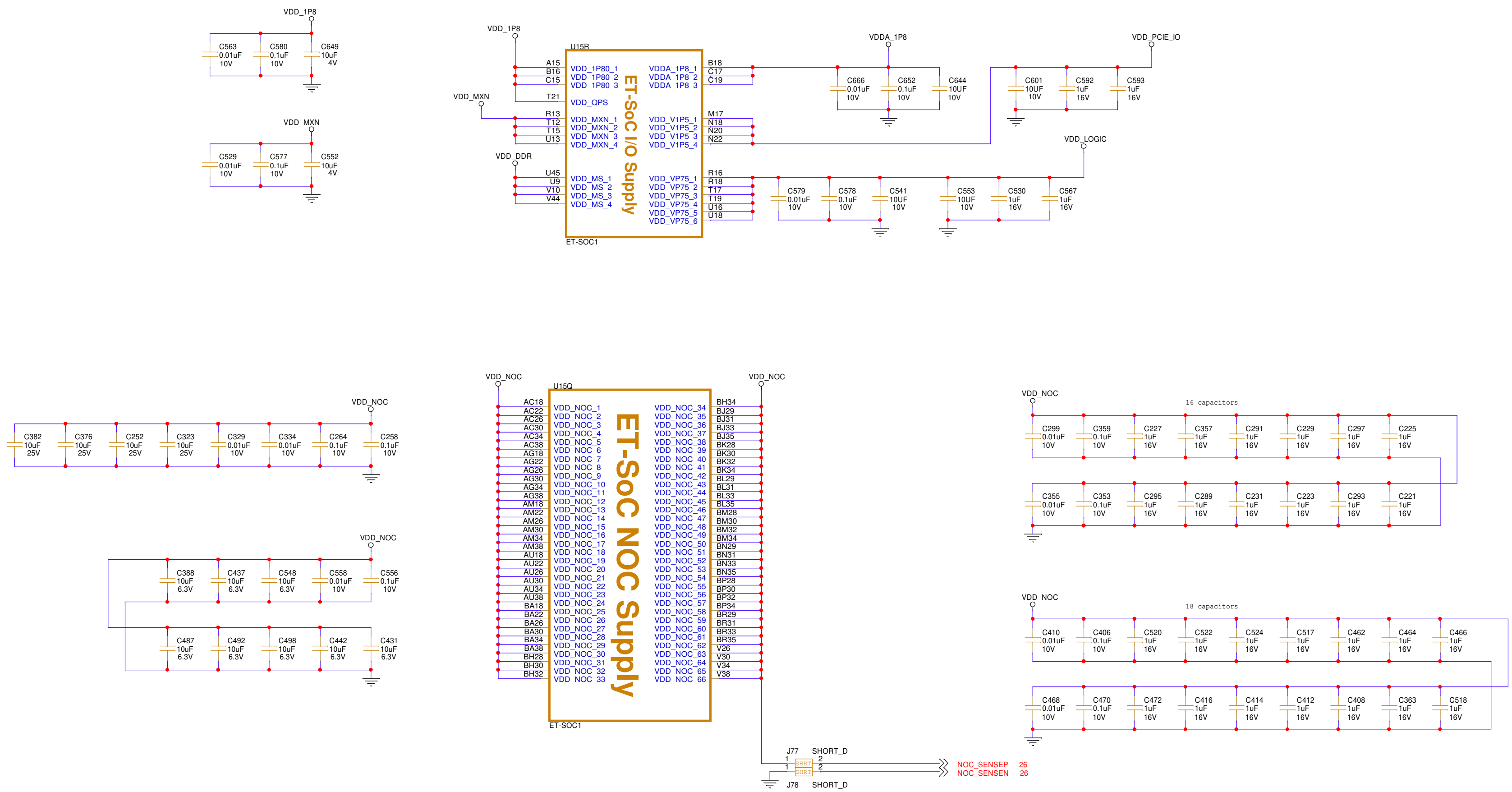

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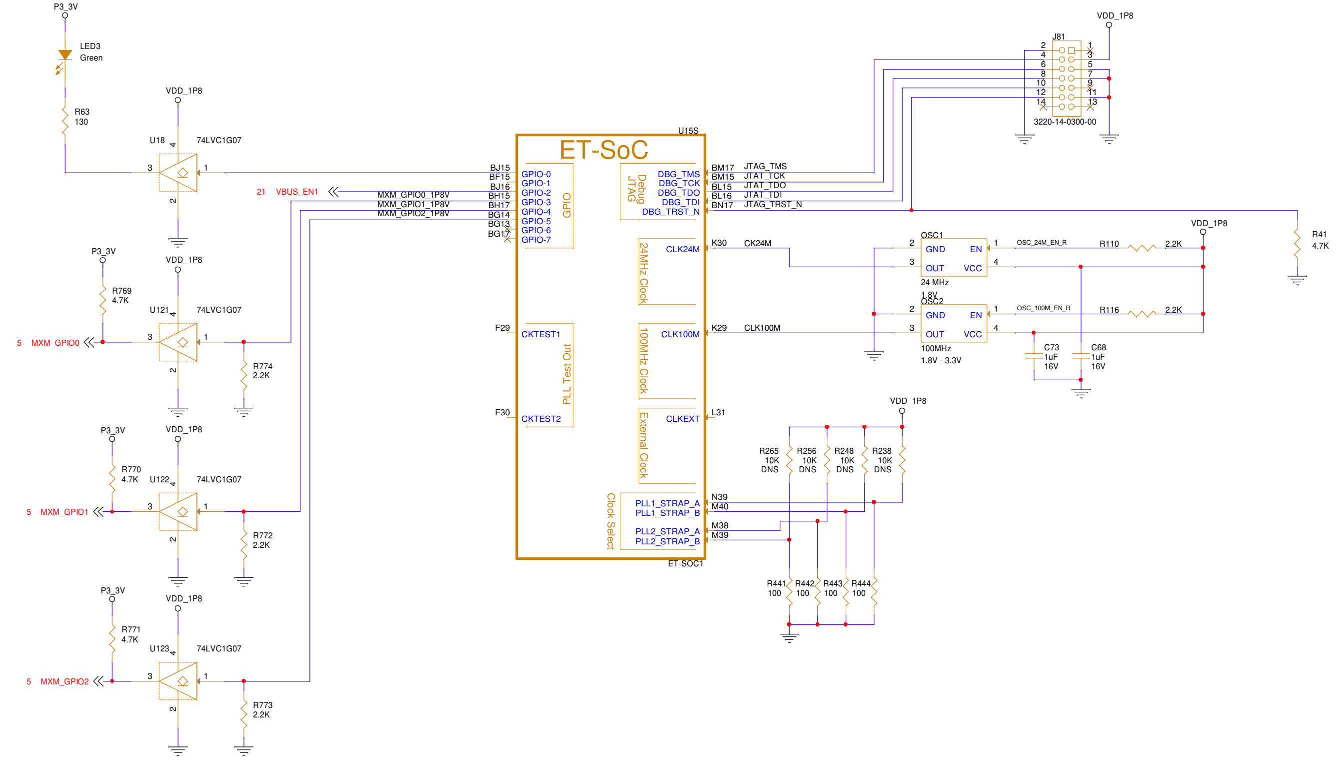
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 18 - SOC - NOC, Maxion Power		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 18 of 32	Rev: 1



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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 19 - SOC - GPIO, CLK, JTAG		Drawn By: Bruce Klein	
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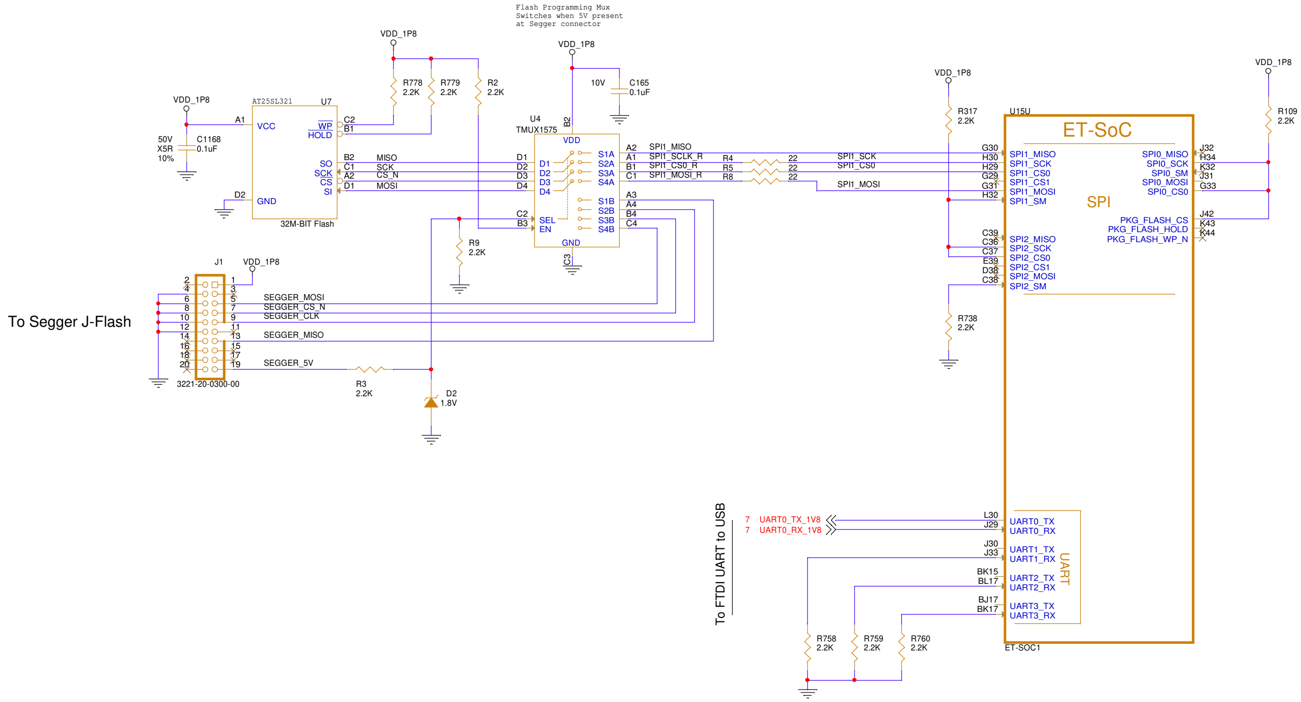

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 20 - SOC - SPI, UART		Drawn By: Bruce Klein	
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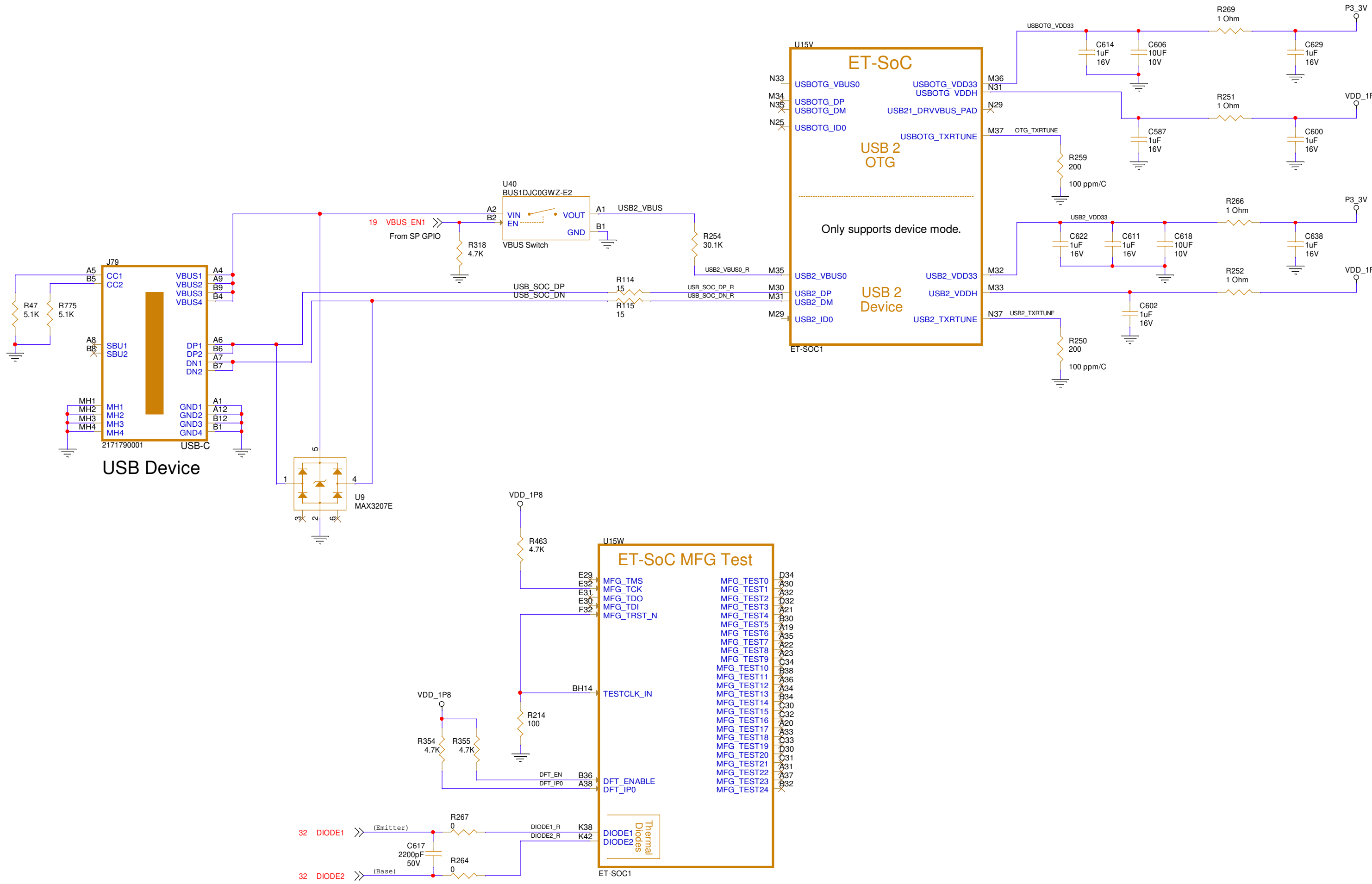

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 21 - SOC - USB/TEST		Drawn By: Bruce Klein	
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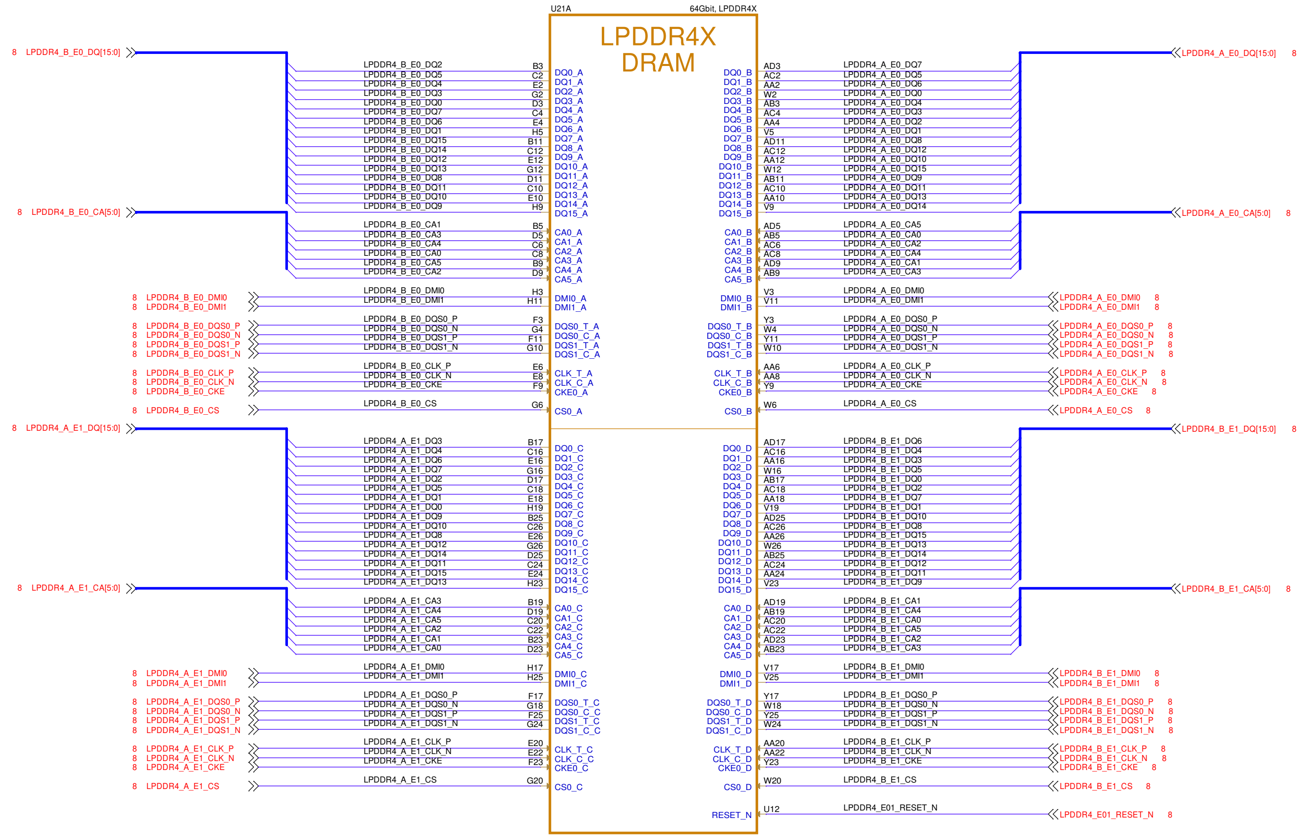

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 22 - LPDDR4x - E01 - 1		Drawn By: Bruce Klein	
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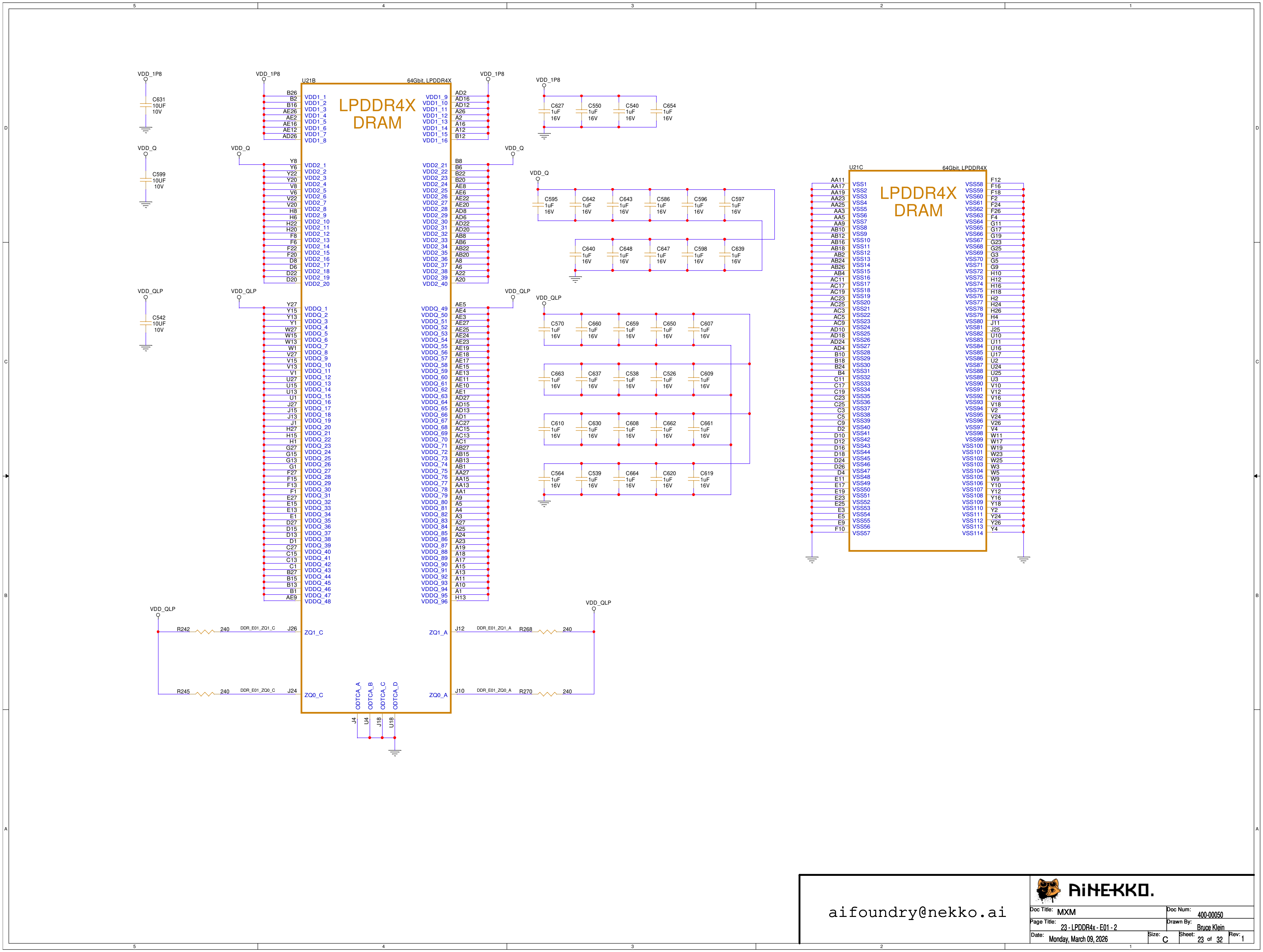

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Doc Title: MXM

Doc Num: 400-00050

Page Title: 23 - LPDDR4x - E01 - 2

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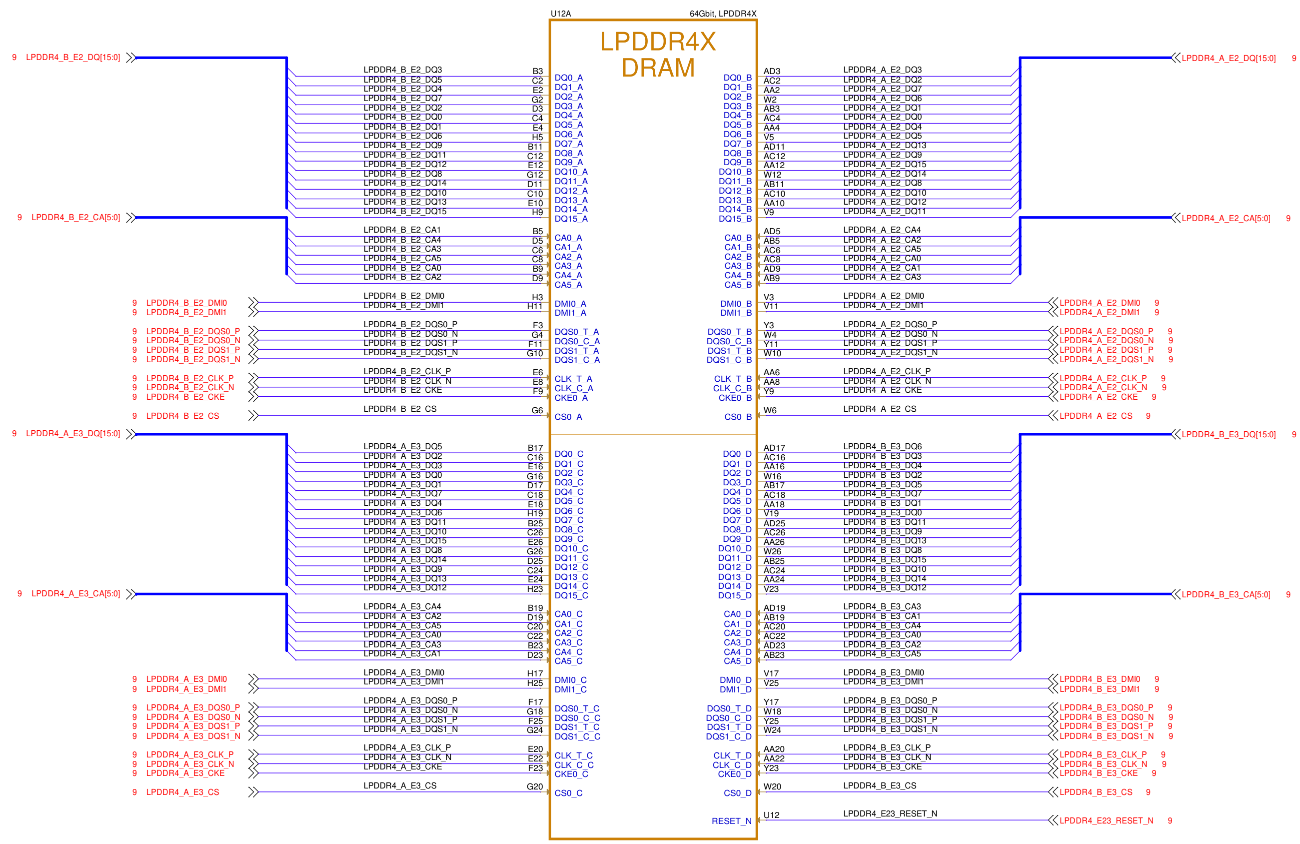

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 24 - LPDDR4x - E23 - 1		Drawn By: Bruce Klein	
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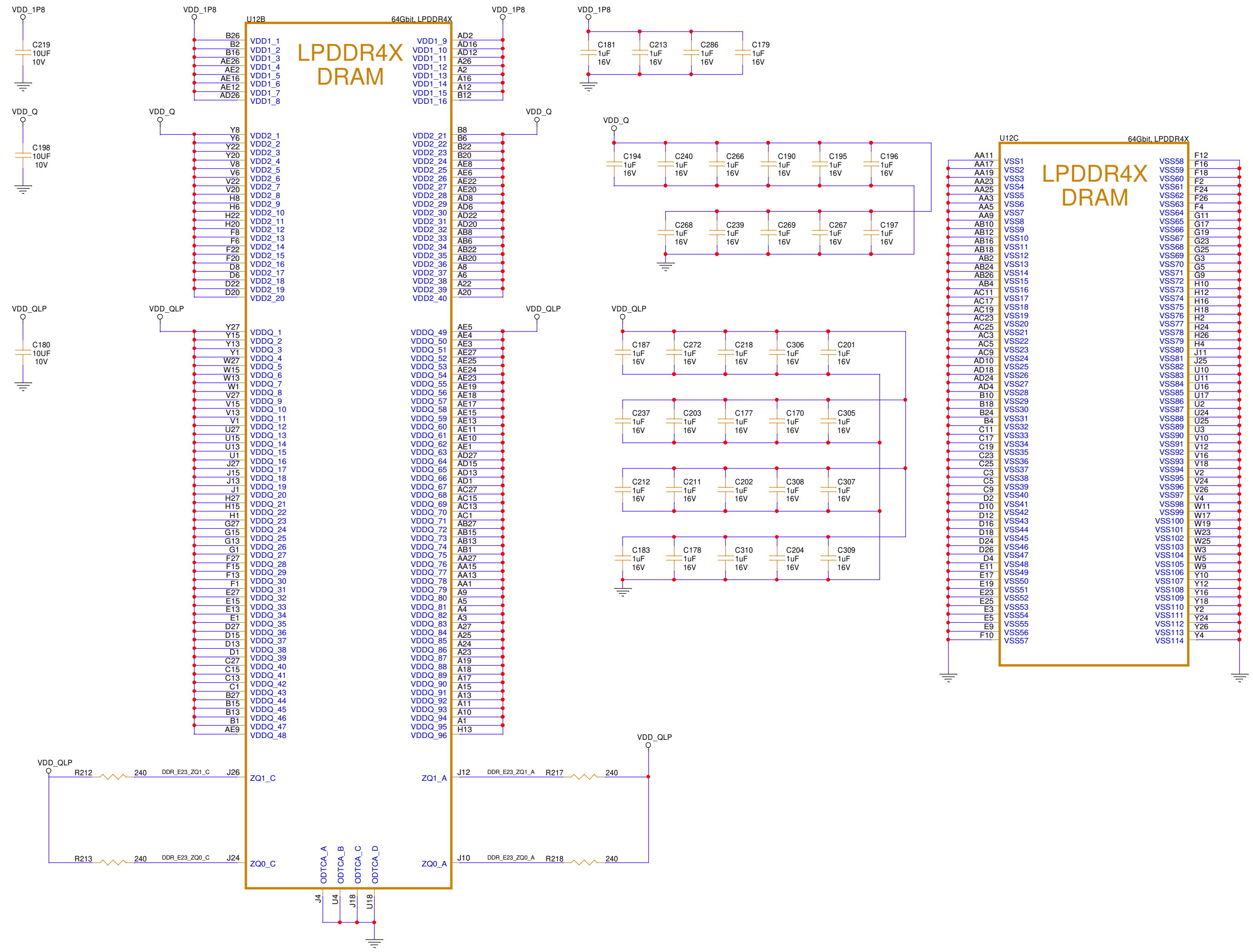

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 25 - LPDDR4x - E23 - 2		Drawn By: Bruce Klein	
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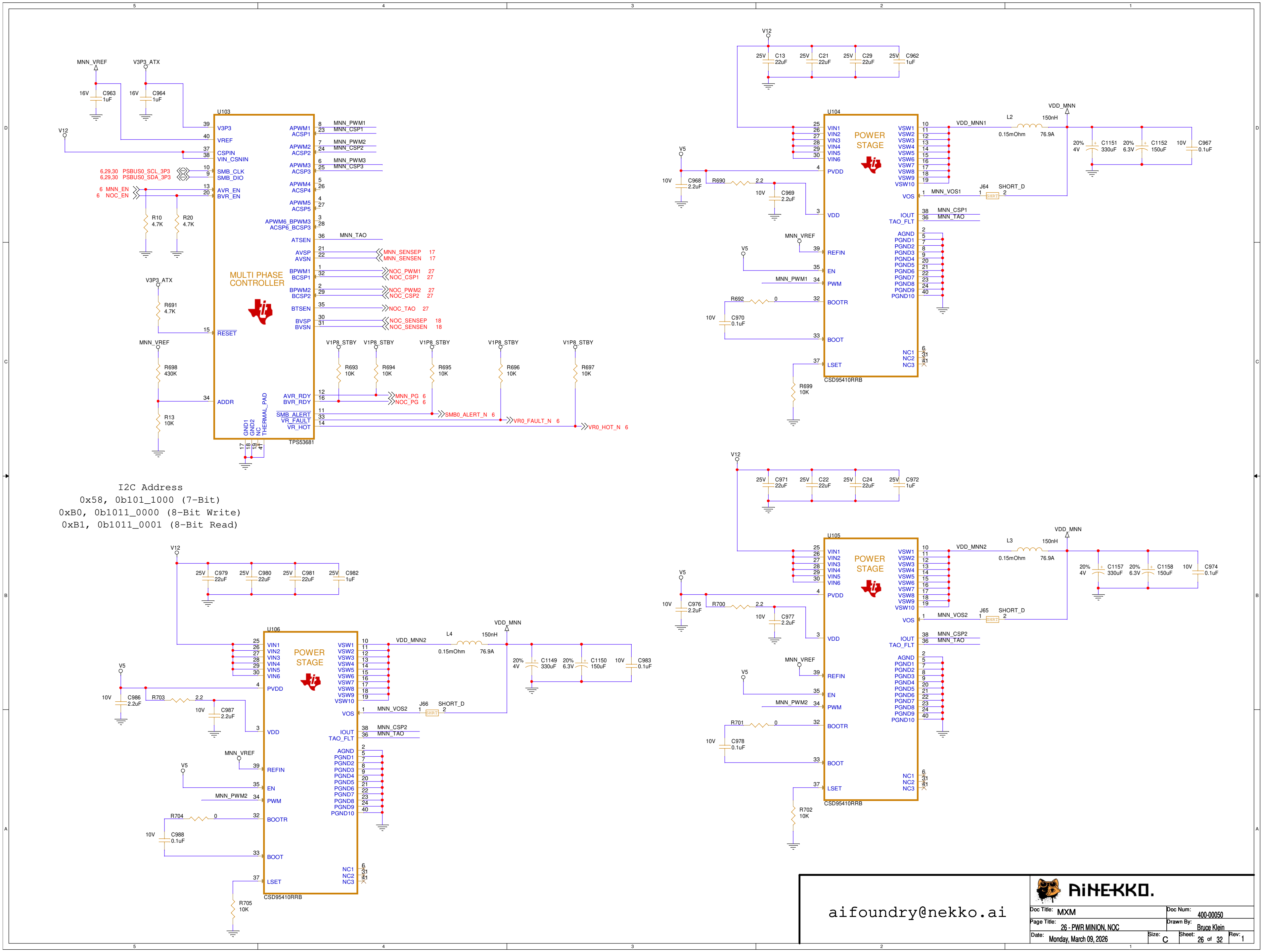

I2C Address
0x58, 0b101_1000 (7-Bit)
0xB0, 0b1011_0000 (8-Bit Write)
0xB1, 0b1011_0001 (8-Bit Read)

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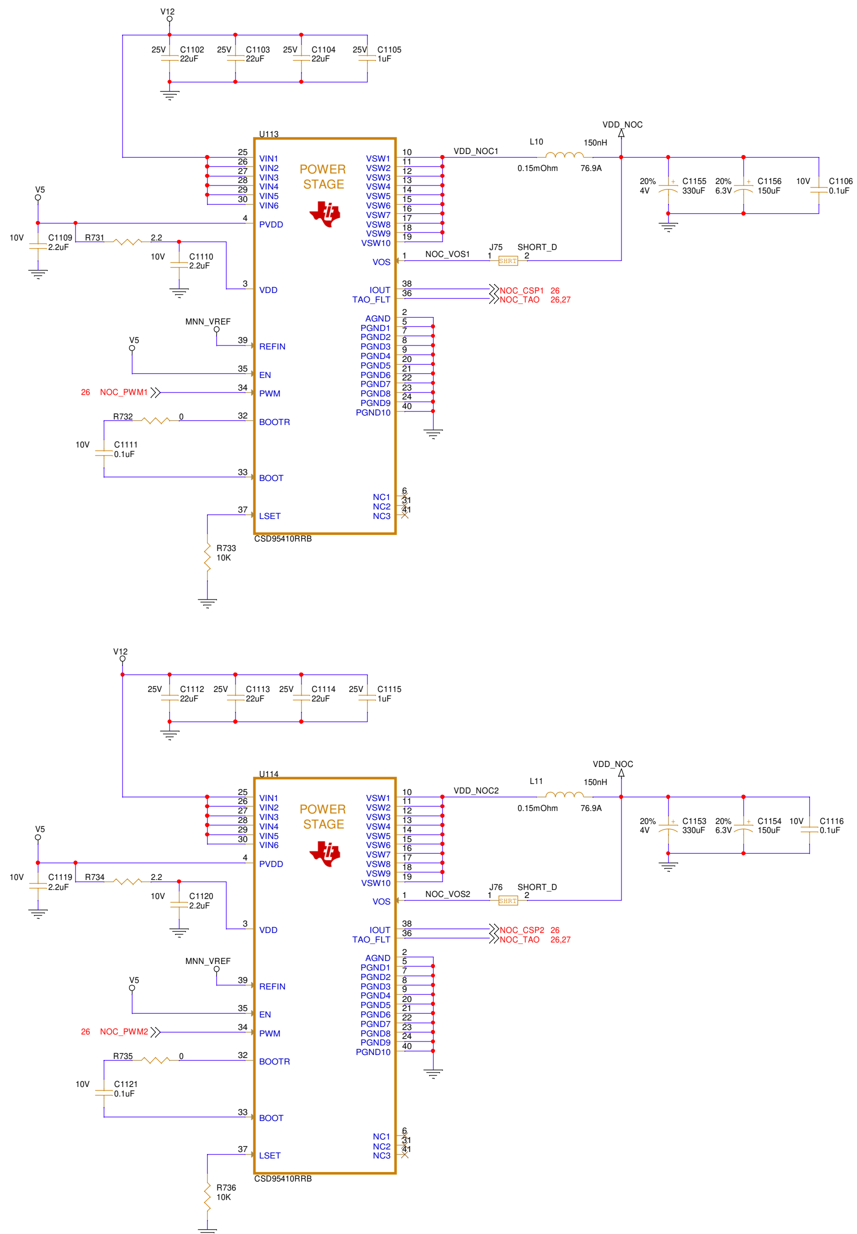
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 26 - PWR MINION, NOC		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 26 of 32	Rev: 1



<Title>

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Doc Title:

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Page Title:

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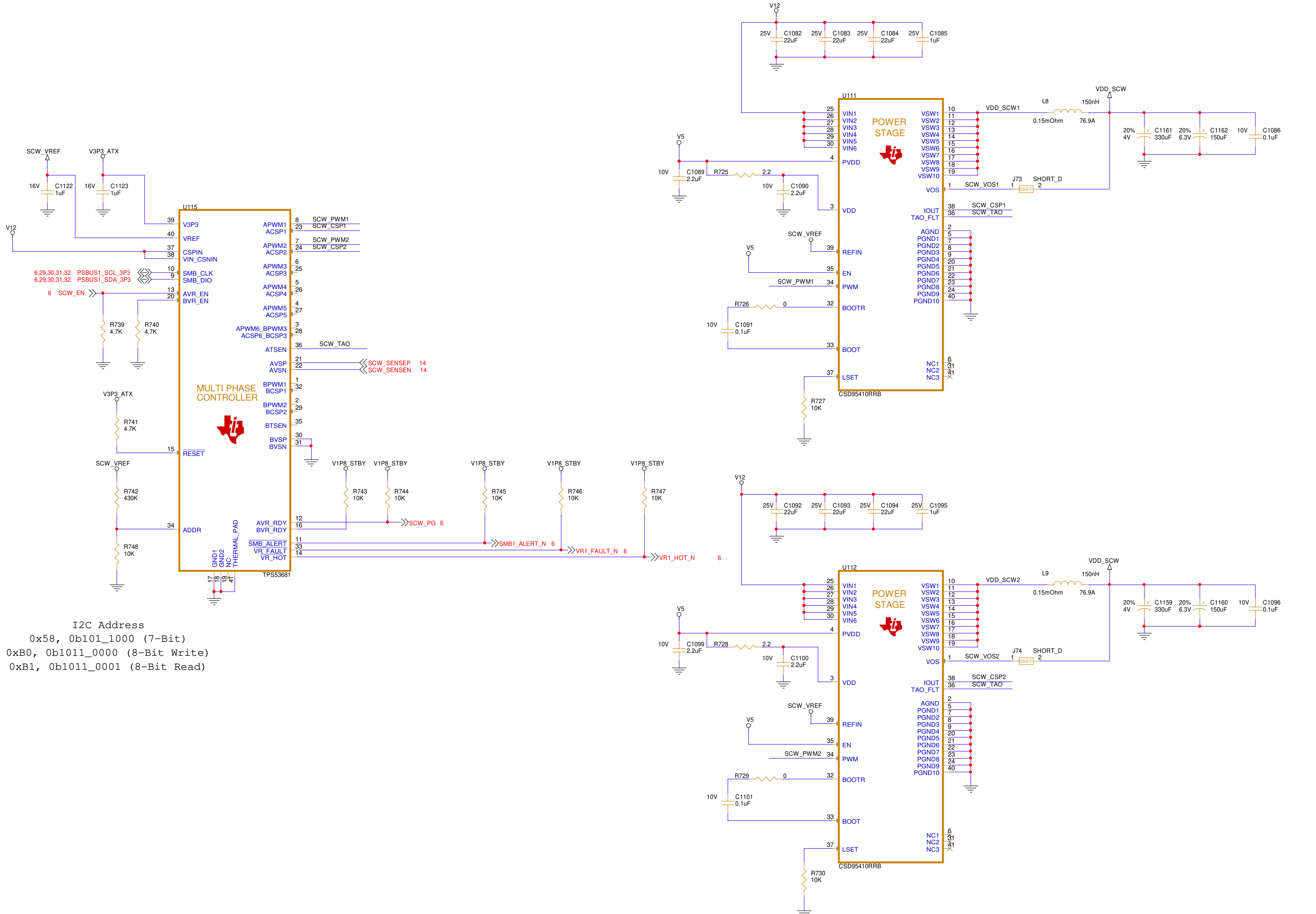

I2C Address
0x58, 0b101_1000 (7-Bit)
0xB0, 0b1011_0000 (8-Bit Write)
0xB1, 0b1011_0001 (8-Bit Read)

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Doc Title: MXM
Doc Num: 400-00050
Page Title: 28 - PWR SCW
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Date: Monday, March 09, 2026
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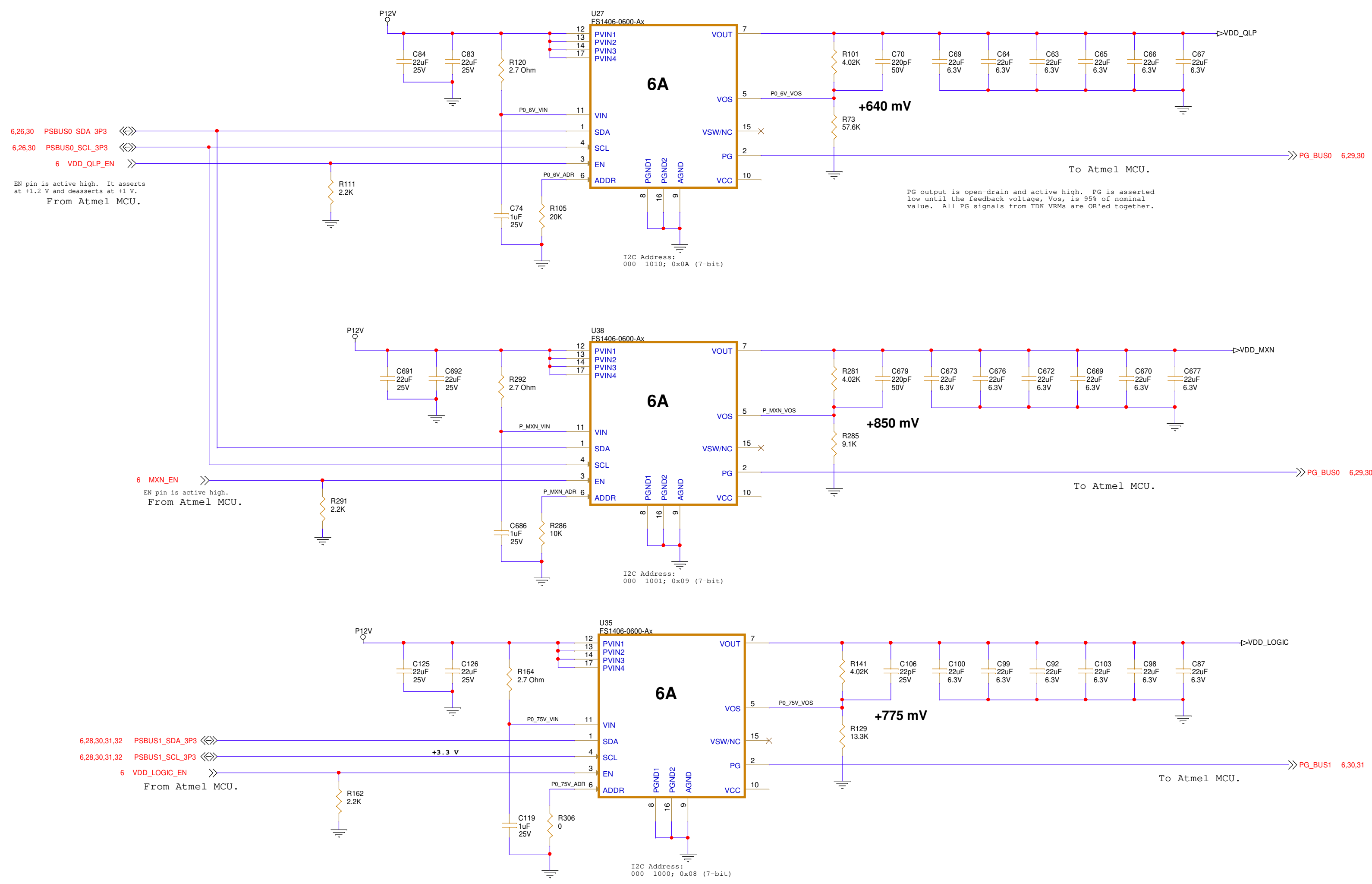

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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 29 - PWR VDD QLP, MXN, LOGIC		Drawn By: Bruce Klein	
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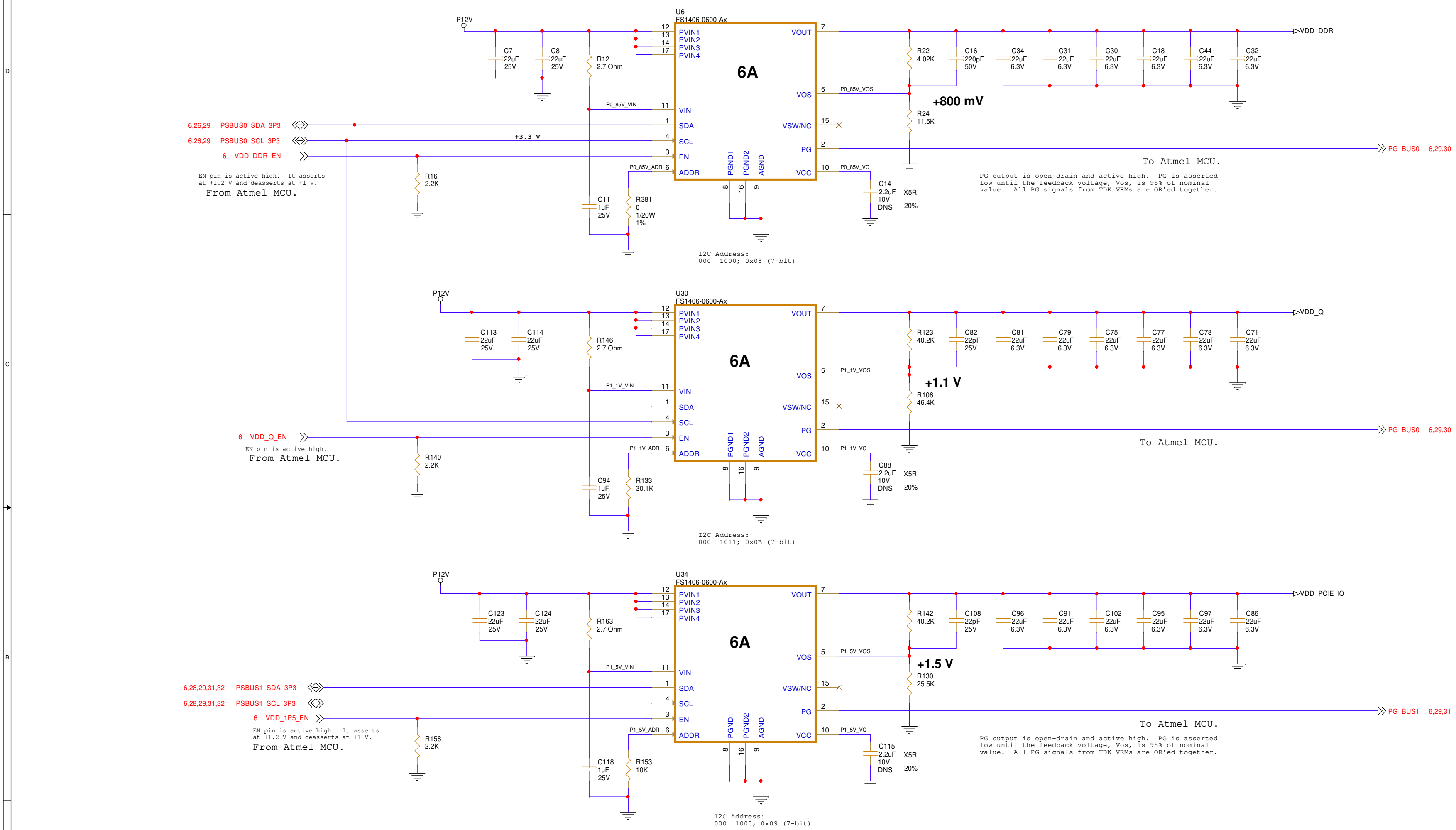

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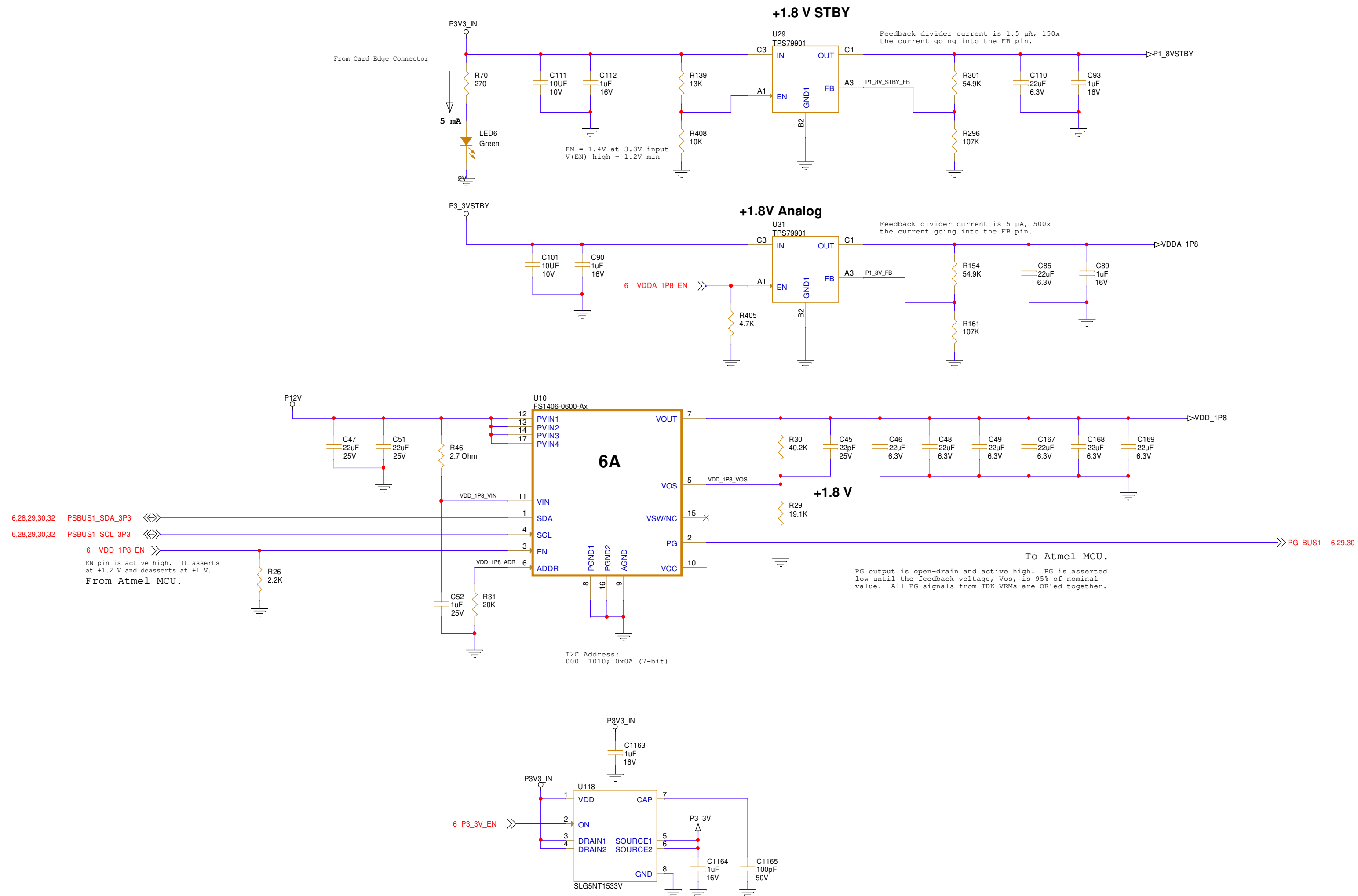
Doc Title: MXM		Doc Num: 400-00050	
Page Title: 30 - PWR VDD DDR, Q, 1P5		Drawn By: Bruce Klein	
Date: Monday, March 09, 2026	Size: C	Sheet: 30 of 32	Rev: 1



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Doc Title: MXM		Doc Num: 400-00050	
Page Title: 31 - PWR 1.8V, 3.3V Switched		Drawn By: Bruce Klein	
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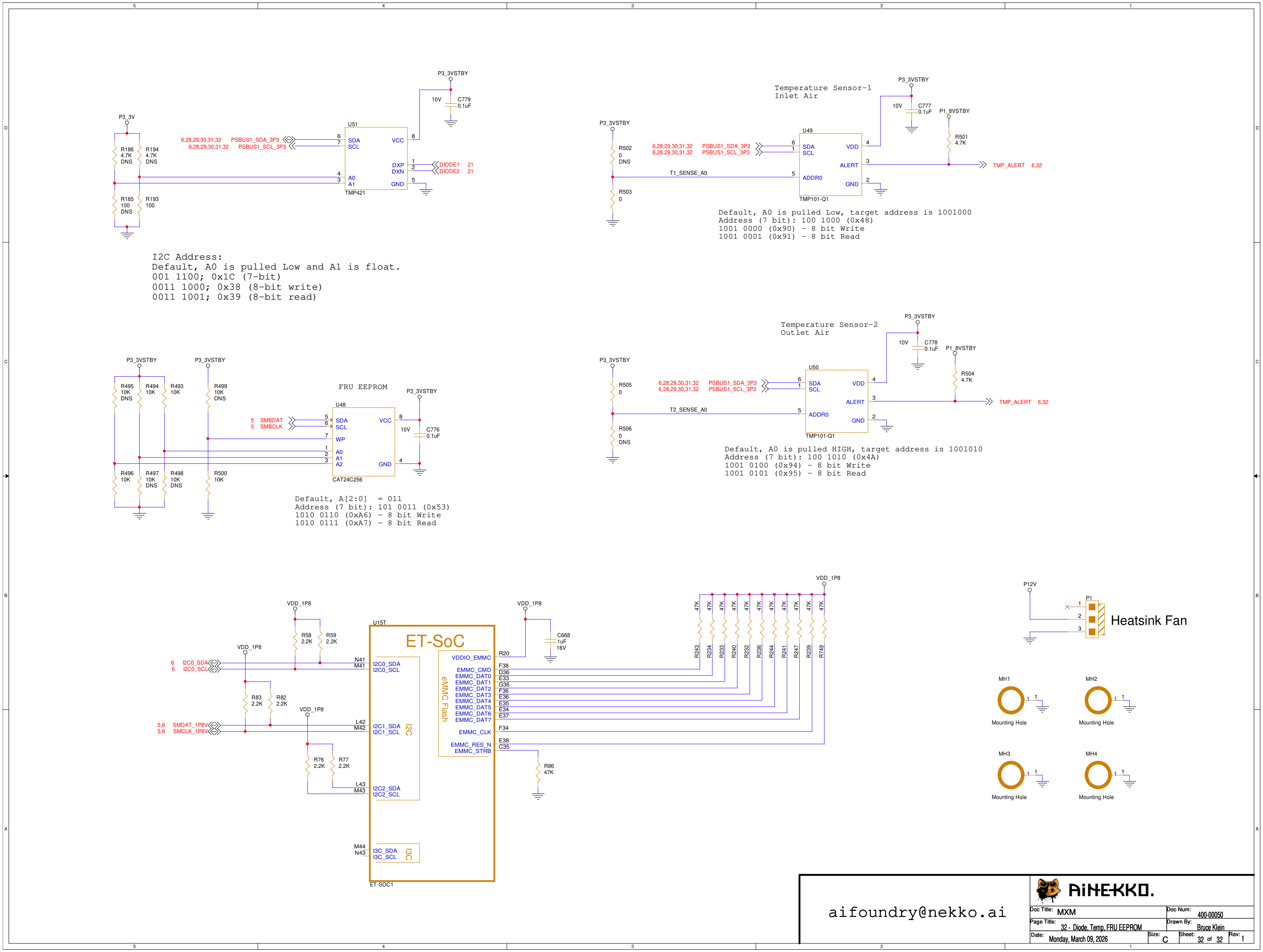

P3_3V
R186 4.7K DNS
R194 4.7K DNS
R185 100 DNS
R193 100
U51
TMP421
SDA
SCL
VCC
DXP
DXN
A0
A1
GND
6,28,29,30,31,32 PSBUS1_SDA_3P3
6,28,29,30,31,32 PSBUS1_SCL_3P3
DIODE1 21
DIODE2 21
P3_3VSTBY
10V
C779 0.1uF
I2C Address:
Default, A0 is pulled Low and A1 is float.
001 1100; 0x1C (7-bit)
0011 1000; 0x38 (8-bit write)
0011 1001; 0x39 (8-bit read)
Temperature Sensor-1
Inlet Air
U49
TMP101-Q1
SDA
SCL
VDD
ALERT
ADDR0
GND
R502 0 DNS
R503 0
T1_SENSE_A0
C777 0.1uF
P1_8VSTBY
R501 4.7K
TMP_ALERT 6,32
Default, A0 is pulled Low, target address is 1001000
Address (7 bit): 100 1000 (0x48)
1001 0000 (0x90) - 8 bit Write
1001 0001 (0x91) - 8 bit Read
Temperature Sensor-2
Outlet Air
U50
TMP101-Q1
R505 0
R506 0 DNS
T2_SENSE_A0
C778 0.1uF
R504 4.7K
TMP_ALERT 6,32
Default, A0 is pulled HIGH, target address is 1001010
Address (7 bit): 100 1010 (0x4A)
1001 0100 (0x94) - 8 bit Write
1001 0101 (0x95) - 8 bit Read
FRU EEPROM
U48
CAT24C256
SDA
SCL
WP
A0
A1
A2
VCC
GND
5 SMBDAT
5 SMBCLK
R495 10K DNS
R494 10K
R493 10K
R499 10K DNS
R496 10K
R497 10K DNS
R498 10K DNS
R500 10K
C776 0.1uF
Default, A[2:0] = 011
Address (7 bit): 101 0011 (0x53)
1010 0110 (0xA6) - 8 bit Write
1010 0111 (0xA7) - 8 bit Read
U15T
ET-SoC
ET-SOC1
VDD_1P8
R58 2.2K
R59 2.2K
R83 2.2K
R82 2.2K
R76 2.2K
R77 2.2K
6 I2C0_SDA
6 I2C0_SCL
5,6 SMDAT_1P8V
5,6 SMCLK_1P8V
I2C0_SDA
I2C0_SCL
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C
I3C_SDA
I3C_SCL
I3C
VDDIO_EMMC
EMMC_CMD
EMMC_DAT0
EMMC_DAT1
EMMC_DAT2
EMMC_DAT3
EMMC_DAT4
EMMC_DAT5
EMMC_DAT6
EMMC_DAT7
EMMC_CLK
EMMC_RES_N
EMMC_STRB
eMMC Flash
C668 1uF 16V
R86 47K
R243 R234 R233 R240 R232 R236 R244 R241 R247 R239 R749
47K
P12V
P1
Heatsink Fan
MH1
MH2
MH3
MH4
Mounting Hole
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Doc Title: MXM
Doc Num: 400-00050
Page Title: 32 - Diode, Temp, FRU EEPROM
Drawn By: Bruce Klein
Date: Monday, March 09, 2026
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